

Lab02

```
library(readr)
library(dplyr)
```

Anexando pacote: 'dplyr'

Os seguintes objetos são mascarados por 'package:stats':

filter, lag

Os seguintes objetos são mascarados por 'package:base':

intersect, setdiff, setequal, union

1)Quais são as estatísticas suficientes para a determinação do percentual de vôos atrasados na chegada (ARRIVAL_DELAY > 10)?

RESPOSTA: percentual = voos atrasados/total de voos

2)Crie uma função chamada getStats que, para um conjunto de qualquer tamanho de dados provenientes de flights.csv.zip, execute as seguintes tarefas (usando apenas verbos do dplyr):

```
getStats <- function(input, pos) {

  input %>%
    filter(
      AIRLINE %in% c("AA", "DL", "UA", "US"),
      !is.na(ARRIVAL_DELAY)
    ) %>%
    group_by(YEAR, MONTH, DAY, AIRLINE) %>%
    summarise(
      total = n(),
      atrasados = sum(ARRIVAL_DELAY > 10),
      .groups = "drop"
    )

}
```

3.

```
callback <- DataFrameCallback$new(getStats)

stats <- read_csv_chunked(
  "C:/Users/giuli/Downloads/flights.csv/flights.csv",
  callback = callback,
  chunk_size = 100000,
  col_types = cols_only(
    YEAR = col_integer(),
    MONTH = col_integer(),
    DAY = col_integer(),
```

```

    AIRLINE = col_character(),
    ARRIVAL_DELAY = col_double()
  )
)
names(stats)

```

```
[1] "YEAR"      "MONTH"     "DAY"       "AIRLINE"   "total"     "atrasados"
```

```
stats <- bind_rows(stats)
```

4.

```

computeStats <- function(stats) {

  stats %>% group_by(YEAR, MONTH, DAY, AIRLINE) %>%
    summarise(total = sum(total), atrasados = sum(atrasados), .groups = "drop") %>%
    mutate(Cia = AIRLINE, Data = as.Date(paste(YEAR, MONTH, DAY, sep = "-")),

           Perc = atrasados / total) %>%
    select(Cia, Data, Perc)

}
stats <- computeStats(stats)
names(stats)

```

```
[1] "Cia"  "Data" "Perc"
```

5.

```

library(ggcal)
library(ggplot2)

paleta <- scale_fill_gradient(low = "#4575b4", high = "#d73027")

baseCalendario <- function(stats, cia) {

  dados <- stats %>% filter(Cia == cia)

  ggcal(dados$Data, dados$Perc)
}

cAA <- baseCalendario(stats, "AA")

cDL <- baseCalendario(stats, "DL")

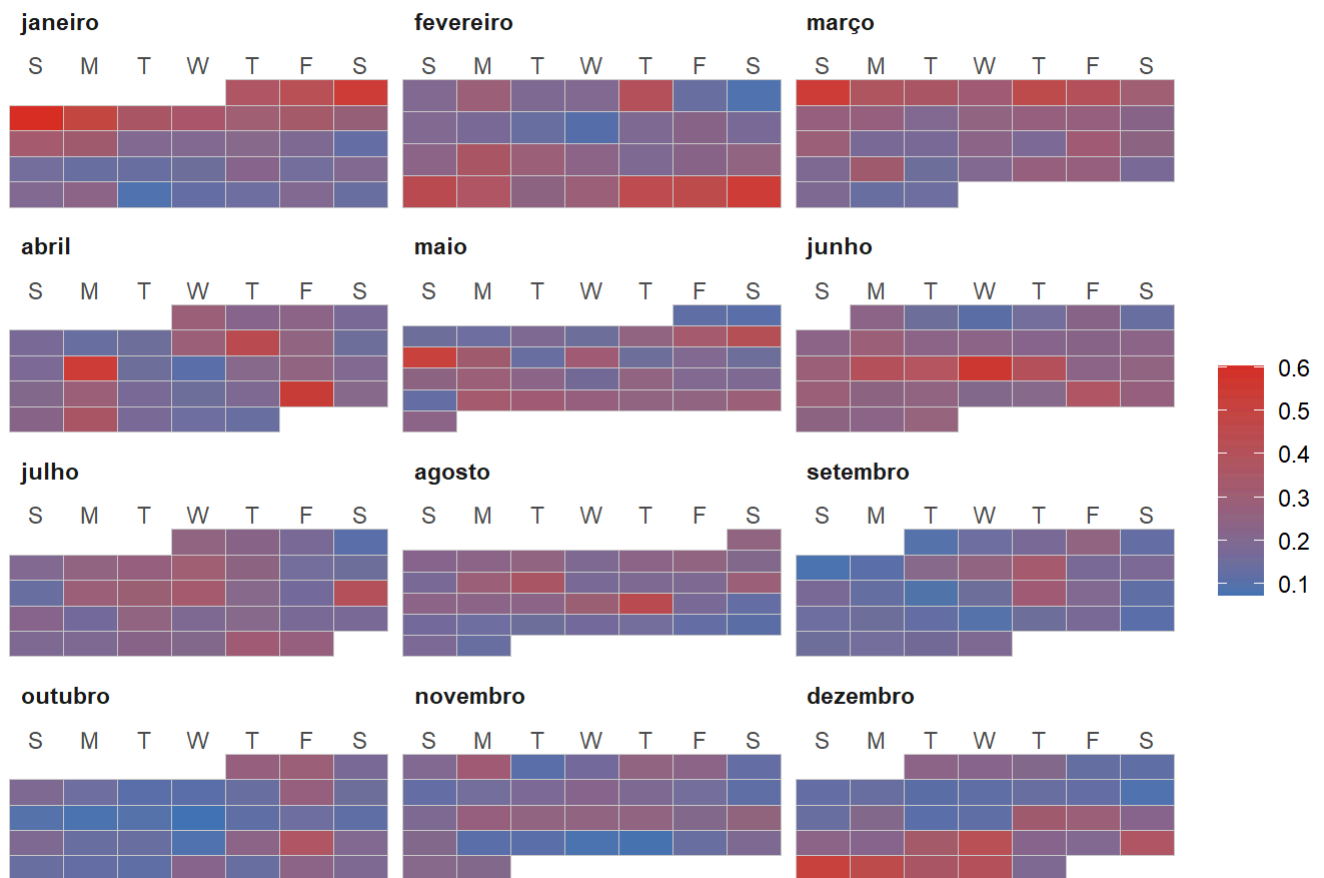
cUA <- baseCalendario(stats, "UA")

cUS <- baseCalendario(stats, "US")

cAA + paleta + ggtitle("Percentual de atrasos - AA")

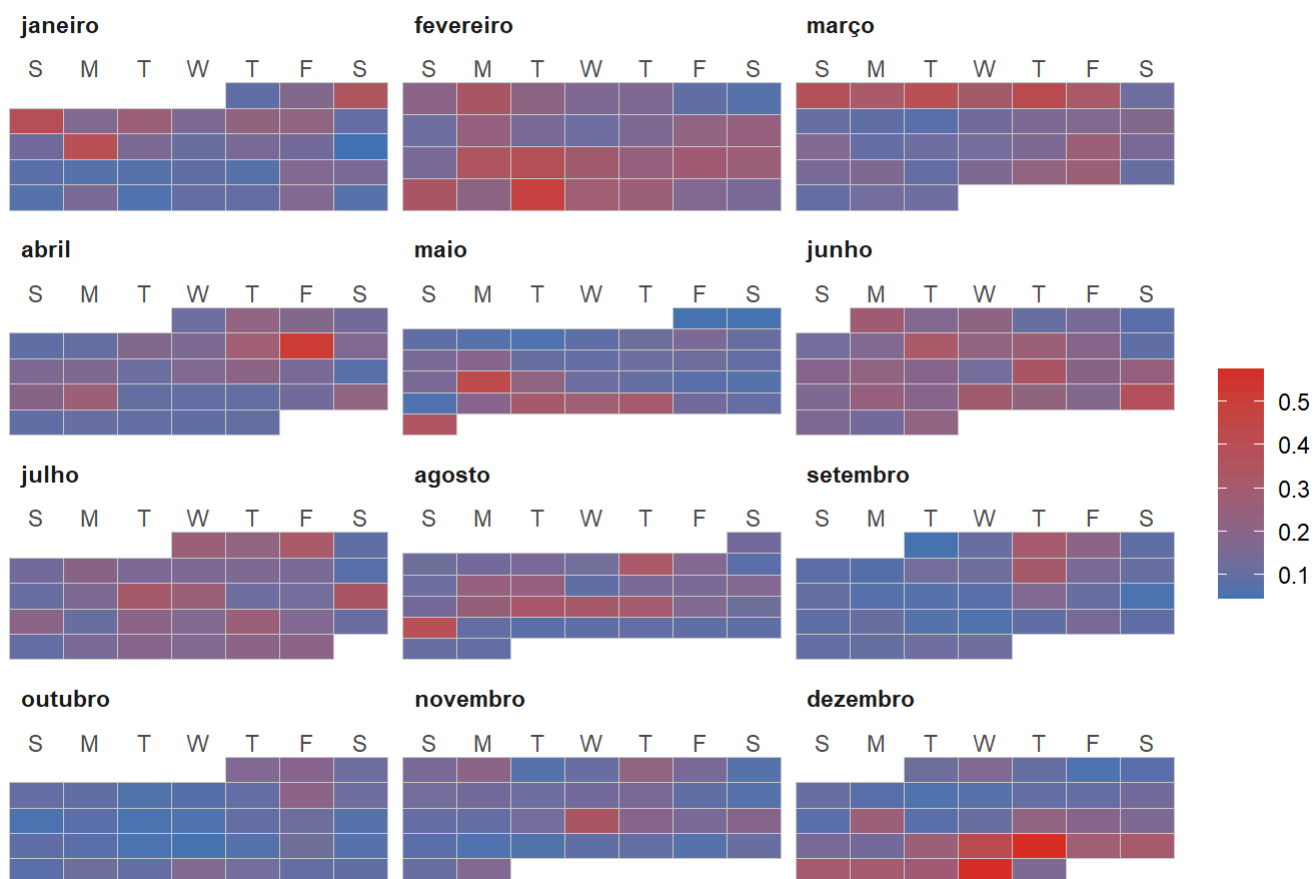
```

Percentual de atrasos - AA



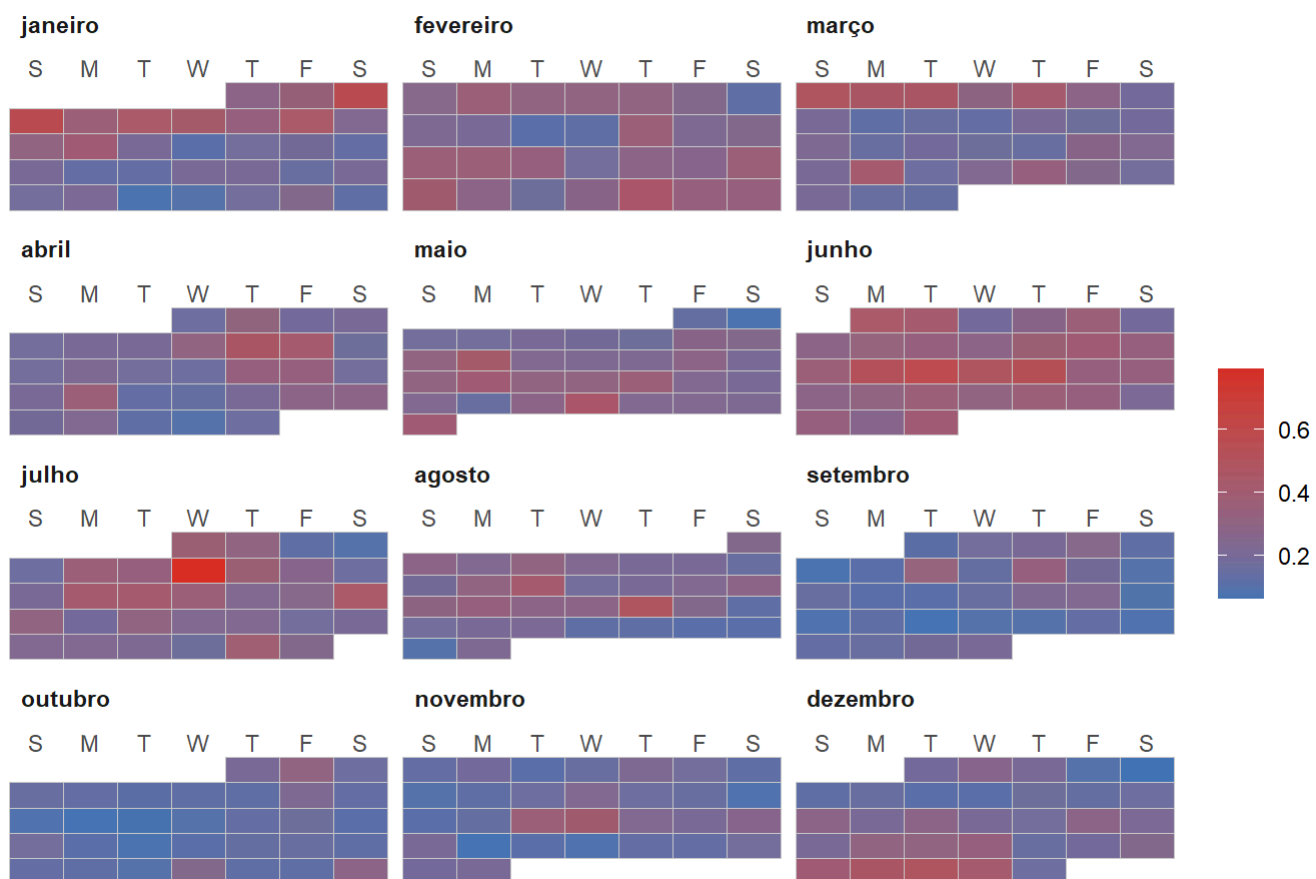
```
cDL + paleta + ggtitle("Delta Airlines")
```

Delta Airlines



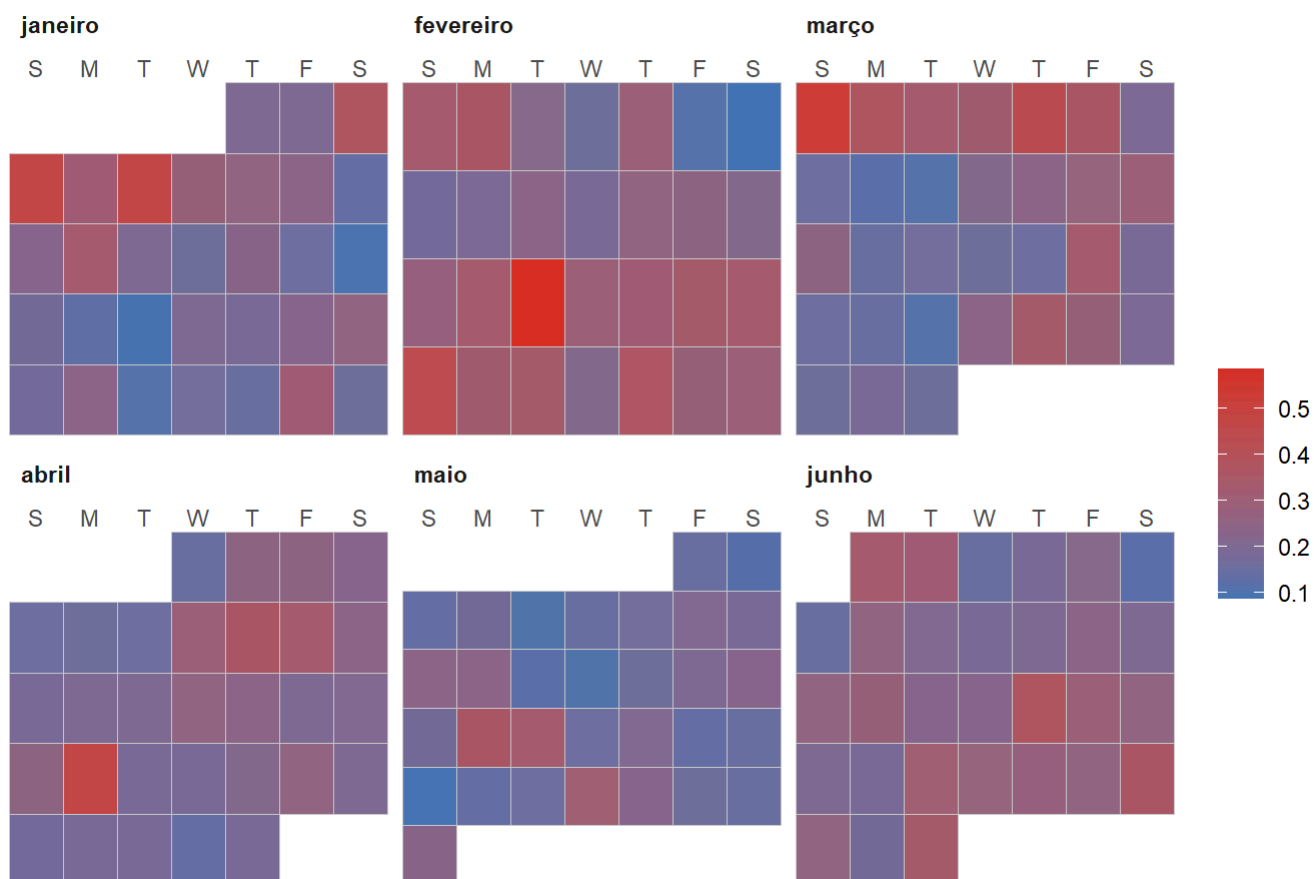
```
cUA + paleta + ggtitle("United Airlines")
```

United Airlines



```
cUS + paleta + ggtitle("US Airways")
```

US Airways



```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-08-25 07:56:10 -03"
```